Dockerizing the QR Code Generator

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GitHub repository
https://github.com/tkg1121/qr-code-generator

DockerHub image
https://hub.docker.com/r/tkg1121/qr-code-generator-app

Successful GitHub Actions run
https://github.com/tkg1121/qr-code-generator/actions/runs/39527047973

Reflection

Containerizing the QR Code Generator helped me connect the Python work from earlier modules with how applications are packaged and shared. The script itself was already provided, so the focus was on building a consistent environment around it. Keeping Python and the required packages inside the image makes it easier to run the application on another computer without repeating the setup manually.

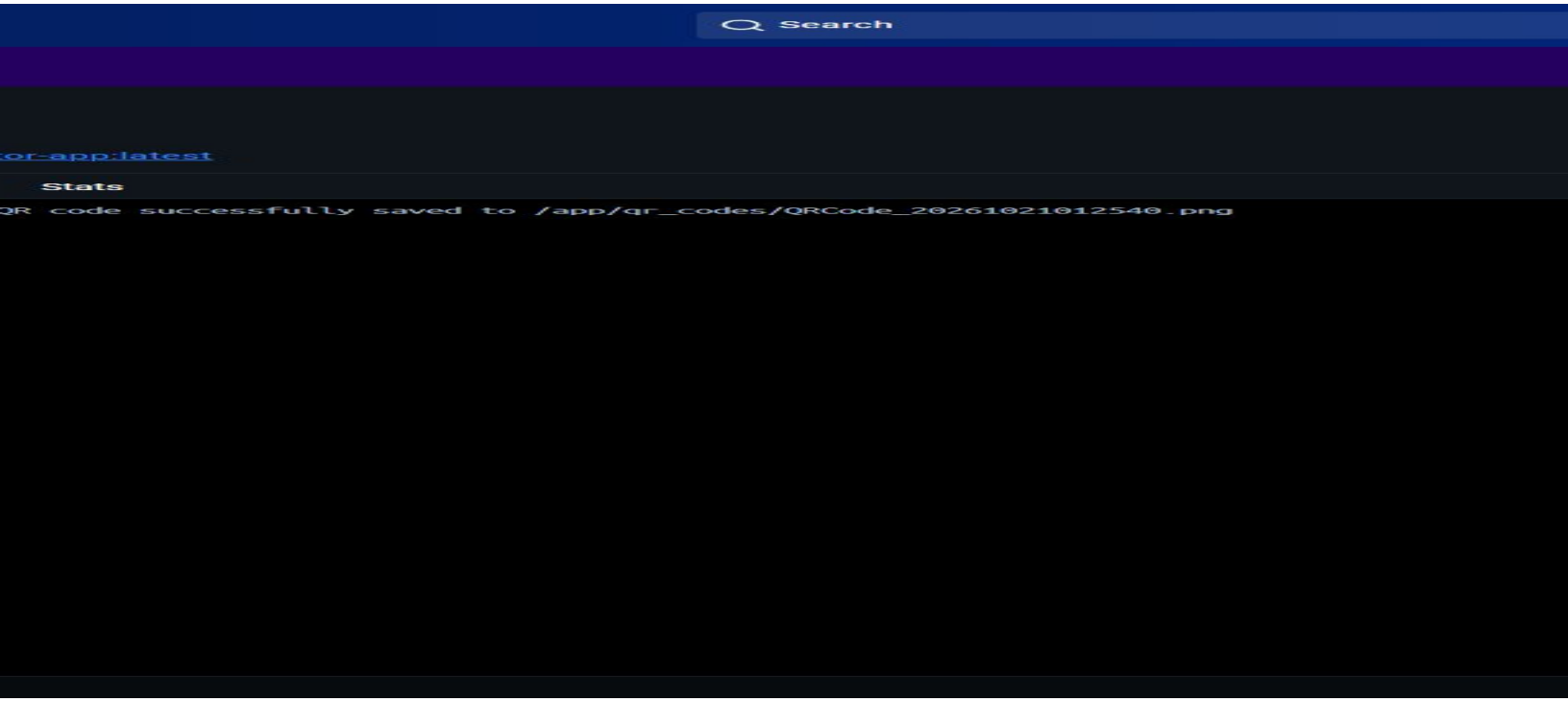
The main part I had to think through was how files move between the container and the host. An image contains the application, but generated QR codes need a volume or bind mount to remain accessible after the container is removed. Running as a non-root user also means the output directory needs the right permissions. Environment variables made it possible to change the output directory and colors without rebuilding the image.

Another useful lesson was understanding the container's lifecycle. This application generates one image and exits, so an Exited (0) status is expected. Checking the logs and decoding the PNG provided better evidence of success than only checking whether a container started. The automated tests and GitHub Actions workflow checked the application, built the image, and verified the container's output.

I feel more comfortable reading a Dockerfile, using ENTRYPOINT and CMD, and pushing an image to DockerHub. I still want more practice with Linux volume permissions and troubleshooting container failures. This assignment also helped clarify the difference between having Docker installed and having its engine running and ready to build images.

Container and Workflow Screenshots

Docker Desktop logs show the QR image was saved successfully. Exited (0) is the expected result for this one-run application.



GitHub Actions passed the automated tests, Docker build, and container output checks.

